

表 6: 不同自监督增强策略的独立作用

数据集	变体	AUROC _{max}	AP _{max}	Δ AUROC	Δ AP
TraceLog	no_ssl	0.8158	0.7291	+0.0000	+0.0000
	edge_perturbation	0.8084	0.7343	-0.0074	+0.0052
	edge_addition	0.8097	0.7309	-0.0061	+0.0018
	node_type_swap	0.8027	0.7284	-0.0131	-0.0007
	edge_type_swap	0.8131	0.7556	-0.0027	+0.0266
	topology_only	0.8079	0.7377	-0.0079	+0.0087
	type_only	0.8201	0.7614	+0.0043	+0.0324
	full	0.8208	0.7426	+0.0050	+0.0135
FlowGraph	all_four	0.8376	0.7778	+0.0217	+0.0487
	no_ssl	0.9996	0.9995	+0.0000	+0.0000
	edge_perturbation	0.9996	0.9995	+0.0000	+0.0000
	edge_addition	1.0000	1.0000	+0.0004	+0.0005
	node_type_swap	0.9708	0.9787	-0.0288	-0.0208
	edge_type_swap	0.9629	0.9757	-0.0367	-0.0238
	topology_only	0.9999	0.9999	+0.0003	+0.0004
	type_only	0.3958	0.3649	-0.6038	-0.6346
ADFA-LD	full	0.9828	0.9848	-0.0168	-0.0147
	all_four	1.0000	1.0000	+0.0004	+0.0005
	no_ssl	0.8269	0.8260	+0.0000	+0.0000
	edge_perturbation	0.8443	0.8383	+0.0174	+0.0123
	edge_addition	0.8515	0.8358	+0.0245	+0.0098
	node_type_swap	0.8302	0.8245	+0.0032	-0.0015
	edge_type_swap	0.8316	0.8250	+0.0046	-0.0010
	topology_only	0.8584	0.8444	+0.0314	+0.0184
	type_only	0.8326	0.8259	+0.0056	-0.0001
	full	0.8474	0.8421	+0.0205	+0.0161
	all_four	0.8650	0.8526	+0.0380	+0.0266

说明: Δ AUROC 和 Δ AP 均以同一数据集上的 no_ssl 为参照。表中各变体按该变体已完成运行的最大 AUROC/AP 报告; all_four 行直接采用主表 HRA-GNN 最佳运行, 以保证完整方法口径与主表一致。